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Ring Theory

- Let R be an integral domain and $p \in R$. If $\langle p\rangle$ is maximal then p is irreducible.
- I is a maximal ideal $\Leftrightarrow R/I$ is a field.
  - Let $a \in R - I$. Then $aR + I = R \Rightarrow 1 \in ab + I$ for some b. So $(a + I)(b + I) = 1 + I$, and every $a \notin I$ has an inverse.
- Let R be an integral domain and $p \in R$. Then $\langle p\rangle$ is prime $\Leftrightarrow p$ is prime.
- Let R be a ring. Then I is prime $\Leftrightarrow R/I$ is an integral domain.
  - $(a + I)(b + I) = I \Rightarrow a$ or $b \in I$
- Maximal ideals are prime.
- Finite integral domains are fields.

Prime Ideals

$\mathbb{Z}_K/\mathfrak{p}$ is finite (lemma 5.20)

Let $\mathfrak{p}$ be a non-zero prime ideal in $\mathbb{Z}_K$. Let $\alpha \in \mathfrak{p}, \alpha \neq 0$. Then $N(\alpha) \in \mathbb{Z}$ and $\alpha | N(\alpha) \Rightarrow N(\alpha) \in \mathfrak{p}$.

$\mathbb{Z}_K$ has integral basis

$$\mathbb{Z}_K = \mathbb{Z}\omega_1 + \cdots + \mathbb{Z}\omega_n$$

Since $N\omega_i \in \mathfrak{p}$ by the nature of ideals, then $a_i\omega_i \equiv b_i\omega_i \mod \mathfrak{p}$ where $0 \leq b_i < N$. It could be smaller but we have established an upper bound for b_i, so $\mathbb{Z}_K/\mathfrak{p}$ is finite.

K is a number field. Every non-zero prime ideal $\mathfrak{p} \subseteq \mathbb{Z}_K$ is maximal (proposition 5.21)

Proof:

- Prime ideal $\mathfrak{p} \Rightarrow \mathbb{Z}_K/\mathfrak{p}$ is an integral domain.
- $\mathbb{Z}_K/\mathfrak{p}$ is finite (lemma 5.20).
- Finite integral domain is a field.
- $\mathbb{Z}_K/\mathfrak{p}$ is a field $\Rightarrow \mathfrak{p}$ is a maximal ideal.

Fractional Ideals

There are prime ideals $\mathfrak{p}_1, ..., \mathfrak{p}_r$ such that $\mathfrak{p}_1 \cdots \mathfrak{p}_r \subseteq \mathfrak{a}$ (lemma 5.24)

$\mathfrak{a}$ is a non-zero ideal of $\mathbb{Z}_K$.

$\mathbb{Z}_K$ is Noetherian. Since $\mathfrak{a}$ forms an ascending chain $\mathfrak{a} \subseteq \mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq \cdots$, it eventually terminates.

There are no prime ideals $\mathfrak{p}_1 \cdots \mathfrak{p}_r \subseteq \mathfrak{a}$. The same is true for all ideals in the chain $\mathfrak{a}_i$.

Lets take $\mathfrak{a}$ to be the largest ideal in the chain.

$\mathfrak{a}$ is not prime otherwise $\mathfrak{p}_1 = \mathfrak{a} \subseteq \mathfrak{a}$ and the proof is finished.

So there are ideals $\mathfrak{a}_1, \mathfrak{a}_2$ in $\mathbb{Z}_K$ such that $\mathfrak{a}_1\mathfrak{a}_2 \subseteq \mathfrak{a}, \mathfrak{a}_1 \notin \mathfrak{a}, \mathfrak{a}_2 \notin \mathfrak{a}$ Write

$$\mathfrak{b}_1 = \mathfrak{a} + \mathfrak{a}_1, \mathfrak{b}_2 = \mathfrak{a} + \mathfrak{a}_2$$

Then we can see that

$$\mathfrak{b}_1\mathfrak{b}_2 = (\mathfrak{a} + \mathfrak{a}_1)(\mathfrak{a} + \mathfrak{a}_2) = \mathfrak{a} + \mathfrak{a}_1\mathfrak{a} + \mathfrak{a}_2\mathfrak{a} + \mathfrak{a}_1\mathfrak{a}_2$$

Since $\mathfrak{a}_1\mathfrak{a}_2 \subseteq \mathfrak{a}$, so $\mathfrak{b}_1\mathfrak{b}_2 \subseteq \mathfrak{a}$. But also observe that

$$\mathfrak{a} \subsetneq \mathfrak{b}_1, \mathfrak{a} \subsetneq \mathfrak{b}_2$$

Since $\mathfrak{b}_1, \mathfrak{b}_2$ are bigger than $\mathfrak{a}$, then by $\mathfrak{a}$'s maximality, there exist prime ideals $\mathfrak{p}_i$ such that

$$\mathfrak{p}_1 \cdots \mathfrak{p}_s \subseteq \mathfrak{b}_1$$

$$\mathfrak{p}_{s+1} \cdots \mathfrak{p}_t \subseteq \mathfrak{b}_2$$

$$\Rightarrow \mathfrak{p}_1 \cdots \mathfrak{p}_t \subseteq \mathfrak{b}_1\mathfrak{b}_2 \subseteq \mathfrak{a}$$

Which is a contradiction.

$\mathfrak{a} \subseteq \mathfrak{b} \Rightarrow \mathfrak{b}^{-1} \subseteq \mathfrak{a}^{-1}$

Let $\beta \in \mathfrak{b}^{-1}$

$$\beta\mathfrak{b} \subseteq \mathbb{Z}_K$$

but $\mathfrak{a} \subseteq \mathfrak{b} \Rightarrow \beta\mathfrak{a} \subseteq \mathbb{Z}_K$ and so

$$\beta \in \mathfrak{a}^{-1}$$

$\mathfrak{a}^{-1} = \{\alpha \in K : \alpha\mathfrak{a} \subseteq \mathbb{Z}_K\}$ is a fractional ideal (lemma 5.25)

$$\mathfrak{a}^{-1} = \{\alpha \in K : \alpha\mathfrak{a} \subseteq \mathbb{Z}_K\}$$

Let $\gamma \in \mathfrak{a}$ and $\mathfrak{c} = \gamma\mathfrak{a}^{-1}$. Take $i, i' \in \mathfrak{c}$, then $i = \gamma\beta, i' = \gamma\beta'$ with $\beta, \beta' \in \mathfrak{a}^{-1}$.

$$(\beta + \beta')\mathfrak{a} = \beta\mathfrak{a} + \beta'\mathfrak{a} \subseteq (\mathbb{Z}_K + \mathbb{Z}_K) = \mathbb{Z}_K$$

Let $i = \gamma\beta \in \mathfrak{c}$ with $\gamma \in \mathfrak{a}, \beta \in \mathfrak{a}^{-1}$ and $r \in \mathbb{Z}_K$. We want to show that $ri \in \mathfrak{c}$.

But note that $r \in \mathfrak{a}^{-1}$, so $r\beta \in \mathfrak{a}^{-1} \Rightarrow ri = \gamma(r\beta) \in \mathfrak{c}$.

```
sage: K.<a> = NumberField(x^2 + 5)
sage: O = K.ring_of_integers()
sage: I = O.ideal(1 + a)
sage: (1 - a) * I
Fractional ideal (6)
sage: (1 - a)/6 * I
Fractional ideal (1)
sage: 1 - a in I^-1
True
sage: a in I^-1
True
sage: I.basis()
[6, a + 1]
sage: factor(I)
(Fractional ideal (2, a + 1)) * (Fractional ideal (3, a + 1))
```

$\mathfrak{a}$ is a proper ideal of $\mathbb{Z}_K \Rightarrow \mathbb{Z}_K \subsetneq \mathfrak{a}^{-1}$ (lemma 5.26)

$\mathfrak{a} \subseteq \mathfrak{b} \Rightarrow \mathfrak{b}^{-1} \subseteq \mathfrak{a}^{-1}$

Let $\beta \in \mathfrak{b}^{-1}$, then $\beta\mathfrak{b} \subseteq \mathbb{Z}_K$.

But $\mathfrak{a} \subseteq \mathfrak{b} \Rightarrow \beta\mathfrak{a} \subseteq \mathbb{Z}_K$

So $\beta \in \mathfrak{a}^{-1}$.

Section 4.6 shows $\langle 1 - \sqrt{-5} \rangle$ is not prime.

```
sage: K.<a> = NumberField(x^2 + 5)
sage: O = K.ring_of_integers()
sage: I = O.ideal(1 + a)
sage: (1 - a) * I
Fractional ideal (6)
sage: (1 - a)/6 * I
Fractional ideal (1)
sage: 1 - a in I^-1
True
sage: a in I^-1
True
sage: I.basis()
[6, a + 1]
sage: I.is_prime()
False
sage: I.is_maximal()
False
sage: I
Fractional ideal (a + 1)
sage: factor(I)
(Fractional ideal (2, a + 1)) * (Fractional ideal (3, a + 1))
sage: J = O.ideal(2, a + 1)
sage: J.is_prime()
True
sage: 7 + a in J
True
sage:   = O.ideal(7 + a)
sage: factor( )
(Fractional ideal (2, a + 1)) * (Fractional ideal (3, a + 1))^3
sage: J.is_prime(), J.is_maximal()  # of course
(True, True)
sage: O.ideal(3 + a+ 1)^3
Fractional ideal (43*a + 4)
sage:   = (43*a + 4)*(10 + a) # choose any random value from the ideal
sage:   in O.ideal(3 + a+ 1)^3
```

```
True
sage:  in
False
sage: *J
Fractional ideal (277830, 7*a + 150115)
sage:
Fractional ideal (a + 7)
sage: 277830 in
True
sage: 7*a + 150115 in
True
sage: * ^-1*J
Fractional ideal (5145, 7*a + 910)
sage: # which is a subset of Z_K
sage: * ^-1
Fractional ideal (119/2*a + 35/2)
sage: J
Fractional ideal (2, a + 1)
sage: (119/2*a + 35/2)*J
Fractional ideal (5145, 7*a + 910)
sage: # so therefore * ^-1 is a subset of J^-1
sage: J^-1
Fractional ideal (1, 1/2*a + 1/2)
sage: # we can see it consists of all odd halfs of a
sage: # and any integer multiple of 1/2
sage: # which * ^-1 = <119/2*a + 35/2> is a member of
sage: (a + 7)*O
Fractional ideal (a + 7)
sage:
434*a - 175
sage: N.<a> = Integers(5)[]
sage: N(a + 7)
a + 2
sage: N(434*a - 175)
4*a
sage: # so they are different
```

$$\alpha \in \mathfrak{p} \Rightarrow \langle \alpha \rangle \subseteq \mathfrak{p}$$

And there exists

$$\mathfrak{p}_1 \cdots \mathfrak{p}_r \subseteq \langle \alpha \rangle$$

but since r is minimal

$$\mathfrak{p}_2 \cdots \mathfrak{p}_r \nsubseteq \langle \alpha \rangle$$

Let $\beta \in \mathfrak{p}_2 \cdots \mathfrak{p}_r$, then $\beta \notin \langle \alpha \rangle$.

$$\beta\mathfrak{p} \subseteq \mathfrak{p}_1 \cdots \mathfrak{p}_r \implies \beta\mathfrak{p} \subseteq \langle \alpha \rangle$$
$$\alpha^{-1}\beta\mathfrak{p} \subseteq \mathbb{Z}_K$$
$$\alpha^{-1}\beta \in \mathfrak{p}^{-1}$$

But also $\beta \notin \langle \alpha \rangle$

$$\Rightarrow \alpha^{-1}\beta \notin \mathbb{Z}_K$$

$\mathfrak{p}$ is maximal $\Rightarrow \mathfrak{p}\mathfrak{p}^{-1} = \mathbb{Z}_K$ (lemma 5.28)

$\mathfrak{p}^{-1}$ strictly contains $\mathbb{Z}_K$, so there is a non-integer element $\theta \in \mathfrak{p}^{-1}$, and $\mathfrak{p}\theta \nsubseteq \mathfrak{p}$. But $\mathfrak{p}$ is maximal, so $\mathfrak{p}\mathfrak{p}^{-1} = \mathbb{Z}_K$.

$\mathfrak{a}$ is any ideal $\Rightarrow \mathfrak{a}\mathfrak{a}^{-1} = \mathbb{Z}_K$ (lemma 5.29)

By the prev lemma, max ideals $\mathfrak{p}\mathfrak{p}^{-1} = \mathbb{Z}_K$. So $\mathfrak{a}$ is not maximal.

Derive identity

$\mathfrak{a}\mathfrak{p}^{-1}$ is an ideal.

$$\mathfrak{a} \subseteq \mathfrak{a}\mathfrak{p}^{-1}$$

but $\exists \theta \in \mathfrak{p}^{-1} : \theta \notin \mathbb{Z}_K$ so $\mathfrak{a} \subsetneq \mathfrak{a}\mathfrak{p}^{-1}$.

Since $\mathfrak{a}\mathfrak{p}^{-1}$ is an ideal, and $\mathfrak{a}$ is the biggest such that $\mathfrak{a}\mathfrak{a}^{-1} = \mathbb{Z}_K$ then

$$\mathfrak{a}\mathfrak{p}^{-1}(\mathfrak{a}\mathfrak{p}^{-1}) = \mathbb{Z}_K$$

Prove final statement

$$\begin{gathered}\mathfrak{a}\mathfrak{p}^{-1}(\mathfrak{a}\mathfrak{p}^{-1}) = \mathbb{Z}_K \\ [\mathfrak{p}^{-1}(\mathfrak{a}\mathfrak{p}^{-1})] \cdot \mathfrak{a} = \mathbb{Z}_K \\ \Rightarrow \mathfrak{p}^{-1}(\mathfrak{a}\mathfrak{p}^{-1}) \subseteq \mathfrak{a}^{-1}\end{gathered}$$

by the definition of a fractional ideal.

$$\Rightarrow \mathfrak{a}\mathfrak{p}^{-1}(\mathfrak{a}\mathfrak{p}^{-1}) \subseteq \mathfrak{a}\mathfrak{a}^{-1}$$

Every ideal $\mathfrak{a} \neq 0$ is a product of prime ideals (lemma 5.31)

Every maximal ideal is prime.

Let $\mathfrak{a}$ be the biggest ideal not a product of primes. Then it is contained in $\mathfrak{p}$ prime and so we can write.

$$\begin{gathered}\mathfrak{a}\mathfrak{p}^{-1} = \mathfrak{p}_1 \cdots \mathfrak{p}_r \\ \Rightarrow \mathfrak{a} = \mathfrak{p}\mathfrak{p}_1 \cdots \mathfrak{p}_r\end{gathered}$$

Norms of Ideals

$N_{K/\mathbb{Q}}(\langle\alpha\rangle) = |N_{K/\mathbb{Q}}(\alpha)|$ (lemma 5.35)

Index calculated from determinant

See Alaca ANT theorem 9.1.2.

Let G be a free Abelian group with n generators $\omega_1, ..., \omega_n$.

$$G = \{x_1\omega_1 + \cdots + x_n\omega_n : x_i \in \mathbb{Z}\}$$

Let H be a subgroup of G generated by n elements $\eta_1, ..., \eta_n$

$$H = \{y_1\eta_1 + \cdots + y_n\eta_n : y_i \in \mathbb{Z}\}$$

Because each $\eta_i \in H \subseteq G$ we have

$$\eta_i = c_{i,1}\omega_1 + \cdots + c_{i,n}\omega_n$$

Let $C = (c_{i,j})$ be an $n \times n$ matrix. Then

$$[G : H] = \begin{cases} |\det(C)| & \text{if } \det(C) \neq 0 \\ \infty & \text{if } \det(C) = 0 \end{cases}$$

where $|\det(C)|$ means absolute value of C's determinant.

Elements of ideal for $\langle\alpha\rangle$

$$\langle\alpha\rangle = \mathbb{Z}\alpha\omega_1 + \cdots + \mathbb{Z}\alpha\omega_n$$

$$\begin{gathered}\alpha\omega_1 = a_{1,1} + \cdots + a_{n,1}\omega_n \\ \alpha\omega_2 = a_{1,2} + \cdots + a_{n,2}\omega_n \\ \cdots \\ \alpha\omega_n = a_{1,n} + \cdots + a_{n,n}\omega_n\end{gathered}$$

$$\alpha \begin{pmatrix} \omega_1 \\ \vdots \\ \omega_n \end{pmatrix} = \begin{pmatrix} a_{1,1} & \dots & a_{n,1} \\ & \dots & \\ a_{1,n} & \dots & a_{n,n} \end{pmatrix} \begin{pmatrix} \omega_1 \\ \vdots \\ \omega_n \end{pmatrix}$$

The definition of norm from 3.2, is given as the determinant of that transform matrix.

$N(\mathfrak{ab}) = N(\mathfrak{a})N(\mathfrak{b})$ (theorem 5.37)

Let $\mathfrak{p}$ be a non-zero prime ideal of $\mathbb{Z}_K$.

$\mathbb{Z}_K/\mathfrak{p} \cong \mathfrak{a}/\mathfrak{ap}$ (lemma 5.36)

There is no ideal $\mathfrak{b}$ between $\mathfrak{ap} \subsetneq \mathfrak{b} \subsetneq \mathfrak{a}$. To see this simply multiply through by $\mathfrak{a}^{-1}$, and note $\mathfrak{p}$ is maximal. So either $\mathfrak{b} = \mathfrak{a}$ or $\mathfrak{ap}$.

Choose $\alpha \in \mathfrak{a}$ with $\alpha \notin \mathfrak{ap}$. Then because of above $\langle \alpha, \mathfrak{ap} \rangle = \mathfrak{a}$.

$$\phi : \mathbb{Z}_K \to \mathfrak{a}/\mathfrak{ap}$$

$$\phi(x) = \alpha x + \mathfrak{ap}$$

is surjective. The kernel is $\langle \mathfrak{p} \rangle$ since $\alpha\langle \mathfrak{p} \rangle = \mathfrak{ap}$.

The book has a typo on the last line of the proof. It should be $\mathbb{Z}_K/\mathfrak{p} \cong \mathfrak{a}/\mathfrak{ap}$.

Result

Factorise $\mathfrak{b}$ into prime ideals and so we just deal with $\mathfrak{b} = \mathfrak{p}$.

$$\phi : \mathbb{Z}_K/\mathfrak{ap} \to \mathbb{Z}_K/\mathfrak{a}$$
$$\phi(\alpha + \mathfrak{ap}) = \alpha + \mathfrak{a}$$

is a homomorphism. So

$$\left| \frac{\mathbb{Z}_K/\mathfrak{ap}}{\mathfrak{a}/\mathfrak{ap}} \right| = \left| \frac{\mathbb{Z}_K/\mathfrak{ap}}{\mathbb{Z}_K/\mathfrak{p}} \right| = |\mathbb{Z}_K/\mathfrak{a}|$$
$$\Rightarrow N(\mathfrak{ab}) = |\mathbb{Z}_K/\mathfrak{ap}| = |\mathbb{Z}_K/\mathfrak{a}| \cdot |\mathbb{Z}_K/\mathfrak{p}| = N(\mathfrak{a})N(\mathfrak{b})$$

Dimension, Ramification Index and Inertia Degree

$\mathbb{Z}_K$ is $n = [K : \mathbb{Q}]$ dimension vector space. See section 3.4.

$$|\mathbb{Z}_K/\langle p \rangle| = p^n$$

By CRT $\mathbb{Z}_K/\langle p \rangle \cong \mathbb{Z}_K/\mathfrak{p}_1^{e_1} \times \mathbb{Z}_K/\mathfrak{p}_r^{e_r}$.

$$|\mathbb{Z}_K/\mathfrak{p}_i^{e_i}| = N(p_i)^{e_i} = [\mathbb{Z}_K/\mathfrak{p}_i : \mathbb{F}_p]^{e_i} = (p^{f_i})^{e_i}$$

$$n = e_1 f_1 + \cdots + e_r f_r$$

```
sage: # See chapter 3.6.1
sage: y = ( (sqrt(2) + sqrt(6))/2 )
sage: minpoly(y)
x^4 - 4*x^2 + 1
sage: K.<a> = NumberField(x^4 - 4*x^2 + 1)
sage: O = K.ring_of_integers()
sage: I = O.ideal(5)
sage: I
Fractional ideal (5)
sage: factor(I)
(Fractional ideal (a^3 - 5*a + 1)) * (Fractional ideal (a^3 - 5*a - 1))
sage: A, B = O.ideal(a^3 - 5*a + 1), O.ideal(a^3 - 5*a - 1)
```

```
sage: A*B
Fractional ideal (5)
sage: A.ramification_index(), B.ramification_index()
(1, 1)
sage: I.norm()
625
sage: A.norm(), B.norm()
(25, 25)
sage: A.norm() * B.norm()
625
sage: ( ( y^3 - 5*y + 1 )*( y^3 - 5*y - 1) ).expand()
5
```

Find Change of Basis Matrix

The ideal $\langle \gamma^3 - 5\gamma + 1 \rangle$ consists of elements $a + b\gamma + c\gamma^2 + d\gamma^3 \in \mathbb{Z}_K$ multiplied by $\gamma^3 - 5\gamma + 1$.

$$\mathbb{Z}_K = \mathbb{Z}[\gamma], \qquad \gamma = \frac{\sqrt{2}+\sqrt{6}}{2}, \qquad \alpha = \sqrt{6} - 1$$
$$\mathbb{Z}_K = \mathbb{Z} + \mathbb{Z}\gamma + \mathbb{Z}y^2 + \mathbb{Z}y^3$$
$$\langle \alpha \rangle = \mathbb{Z}\alpha + \mathbb{Z}\alpha\gamma + \mathbb{Z}\alpha\gamma^2 + \mathbb{Z}\alpha\gamma^3$$
$$\sqrt{2} = \gamma^3 - 3\gamma, \qquad \sqrt{3} = \gamma^2 - 2$$
$$\alpha = -\gamma^3 + 5\gamma - 1$$

Elements in $\langle \alpha \rangle$ are of the form

$$(a + b\gamma + c\gamma^2 + d\gamma^3)(-\gamma^3 + 5\gamma - 1)$$
$$\text{minpoly}(\gamma) = \gamma^4 - 4\gamma^2 + 1$$
$$\implies \gamma^4 = 4\gamma^2 - 1$$

```
sage: var("a b c d y")
(a, b, c, d, y)
sage: Y = (sqrt(2) + sqrt(6))/2
sage: y4 = 4*y^2 - 1
sage: (y4.subs({y: Y}) - Y^4).expand()
0
sage: e = ( (a + b*y + c*y^2 + d*y^3)*(-y^3 + 5*y - 1) ).expand(); e
-d*y^6 - c*y^5 - b*y^4 + 5*d*y^4 - a*y^3 + 5*c*y^3 - d*y^3 + 5*b*y^2 - c*y^2 + 5*a*y - b*y - a
sage: e = e.subs({y^4: y4}).expand(); e
-d*y^6 - c*y^5 - a*y^3 + 5*c*y^3 - d*y^3 + b*y^2 - c*y^2 + 20*d*y^2 + 5*a*y - b*y - a + b - 5*d
sage: e = e.subs({y^5: y*y4}).expand(); e
-d*y^6 - a*y^3 + c*y^3 - d*y^3 + b*y^2 - c*y^2 + 20*d*y^2 + 5*a*y - b*y + c*y - a + b - 5*d
sage: e = e.subs({y^6: y^2*y4}).expand(); e
-4*d*y^4 - a*y^3 + c*y^3 - d*y^3 + b*y^2 - c*y^2 + 21*d*y^2 + 5*a*y - b*y + c*y - a + b - 5*d
sage: e = e.subs({y^4: y4}).expand(); e
-a*y^3 + c*y^3 - d*y^3 + b*y^2 - c*y^2 + 5*d*y^2 + 5*a*y - b*y + c*y - a + b - d
sage: e.collect(y)
-(a - c + d)*y^3 + (b - c + 5*d)*y^2 + (5*a - b + c)*y - a + b - d
```

$$\begin{pmatrix} -a+b-d \\ 5a-b+c \\ b-c+5d \\ -a+c-d \end{pmatrix} = \begin{pmatrix} -1 & 1 & 0 & -1 \\ 5 & -1 & 1 & 0 \\ 0 & 1 & -1 & 5 \\ -1 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$$

```
sage: A = matrix([
....:     [-1, 1, 0, -1],
....:     [5, -1, 1, 0],
....:     [0, 1, -1, 5],
....:     [-1, 0, 1, -1]
....: ])
sage: A.determinant()
25
```

Calculate Index From Basis Transformation Matrix

We can perform 2 operations on this change of basis matrix which keep it valid. See Alaca 9.1.2.

Let $G = \{x_1\omega_1 + x_2\omega_2 + x_3\omega_3 + x_4\omega_4 : x_i \in \mathbb{Z}\}$ with a subgroup H defined by a basis

$$\eta_i = c_{i,1}\omega_1 + c_{i,2}\omega_2 + c_{i,3}\omega_3 + c_{i,4}\omega_4$$

We can add and subtract these basis from each other leaving the subgroup H intact. Observe that

$$\{\eta_1, \eta_2, \eta_3, \eta_4\} \text{ and } \{\eta_1, \eta_2 + k\eta_3, \eta_3, \eta_4\}$$

both generate the same group.

There is a slightly more difficult column operation. We simplify notation below. Assume we are swapping columns 2 and 3 of the 2nd row.

$$\begin{aligned}
\eta_2 &= c_1\omega_1 + c_2\omega_2 + c_3\omega_3 + c_4\omega_4 \\
&= (c_1\omega_1 + c_4\omega_4) + c_2\omega_2 + c_3\omega_3 \\
&= (c_1\omega_1 + c_4\omega_4) + (c_2 + c_3)\omega_2 + c_3(\omega_3 - \omega_2) \\
&= c_1\omega_1 + \bar{c}_2\omega_2 + c_3\bar{\omega}_3 + c_4\omega_4
\end{aligned}$$

where

$$\bar{c}_2 = c_2 + c_3, \qquad \bar{\omega}_3 = \omega_3 - \omega_2$$

Which will leave also G unchanged. Now we end up with

$$G = \langle \bar{\omega}_1, \bar{\omega}_2, \bar{\omega}_3, \bar{\omega}_4 \rangle$$

$$H = \langle d_1\bar{\omega}_1, d_2\bar{\omega}_2, d_3\bar{\omega}_3, d_4\bar{\omega}_4 \rangle$$

See the script ch5-degree.sage where we use this method to compute the degree.

Using Ring Isomorphisms

$$\sigma_1 : \begin{cases} \sqrt{2} & \mapsto \sqrt{2} \\ \sqrt{3} & \mapsto \sqrt{3} \\ \sqrt{6} & \mapsto \sqrt{6} \end{cases} \qquad \sigma_2 : \begin{cases} \sqrt{2} & \mapsto -\sqrt{2} \\ \sqrt{3} & \mapsto \sqrt{3} \\ \sqrt{6} & \mapsto -\sqrt{6} \end{cases} \qquad \sigma_3 : \begin{cases} \sqrt{2} & \mapsto \sqrt{2} \\ \sqrt{3} & \mapsto -\sqrt{3} \\ \sqrt{6} & \mapsto -\sqrt{6} \end{cases} \qquad \sigma_4 : \begin{cases} \sqrt{2} & \mapsto -\sqrt{2} \\ \sqrt{3} & \mapsto -\sqrt{3} \\ \sqrt{6} & \mapsto \sqrt{6} \end{cases}$$

$$\begin{aligned}
N(\sqrt{6} - 1) &= \sigma_1(\sqrt{6} - 1)\sigma_2(\sqrt{6} - 1)\sigma_3(\sqrt{6} - 1)\sigma_4(\sqrt{6} - 1) \\
&= (\sqrt{6} - 1)(-\sqrt{6} - 1)(-\sqrt{6} - 1)(\sqrt{6} - 1) \\
&= 25
\end{aligned}$$

```
sage: R.<x> = PolynomialRing(ZZ, 1)
sage: I = Ideal([x^4 - 4*x^2 + 1, x^3 - 5*x + 1])
sage: I.groebner_basis()
[x^2 + 4*x + 1, 5]
```

Which isomorphic to $\mathbb{F}_{5^2}$.

Deconstructing Primes into Ideals (prop 5.42)

Short Explanation

$$\mathbb{Z}_K/\langle p \rangle \cong \mathbb{F}_p[\gamma] \cong \mathbb{F}_p[X]/\langle \bar{g}(X) \rangle$$

By CRT

$$\mathbb{F}_p[X]/\langle \bar{g}(X) \rangle \cong \mathbb{F}_p[X]/\langle \bar{g}_1(X)^{e_1} \rangle \times \cdots \times \mathbb{F}_p[X]/\langle \bar{g}_r(X)^{e_r} \rangle$$

The map $\mathbb{Z}_K \to \mathbb{F}_p[X]/\langle \bar{g}_1(X)^{e_1} \rangle \times \cdots \times \mathbb{F}_p[X]/\langle \bar{g}_r(X)^{e_r} \rangle$ has kernel

$$\langle p, g_1(\gamma)^{e_1} \rangle \cap \cdots \cap \langle p, g_r(\gamma)^{e_r} \rangle$$

$\mathfrak{p}_i^{e_i} \subseteq \langle p, g_i(\gamma)^{e_i} \rangle$ because

$$\mathfrak{p}_i^{e_i} = \langle p^{e_i}, p^{e_i-1} g_i(\gamma), ..., p g_i(\gamma)^{e_i-1}, g_i(\gamma)^{e_i} \rangle$$

Finally

$$\langle p \rangle = \langle p, g_1(\gamma)^{e_1} \rangle \cap \cdots \cap \langle p, g_r(\gamma)^{e_r} \rangle$$
$$\Rightarrow \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r} \subseteq \langle p \rangle$$

Taking norms, we see that $n = e_1 f_1 + \cdots + e_r f_r$, so the inclusion is actually an equality.

Example

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$$
$$\gamma = \frac{\sqrt{2} + \sqrt{6}}{2}$$
$$g(X) = X^4 - 4X^2 + 1$$
$$p = 5$$

$$\begin{aligned} \bar{g}(X) &= X^4 + X^2 + 1 \\ &= (X^2 + X + 1)(X^2 + 4X + 1) \end{aligned}$$

$$g_1(X) = (X^2 + X + 1), g_2(X) = X^2 + 4X + 1$$
$$\mathfrak{p}_1 = \langle 5, \gamma^2 + \gamma + 1 \rangle, \mathfrak{p}_2 = \langle 5, \gamma^2 + 4\gamma + 1 \rangle$$

Double Quotienting Ideals Isomorphic to Sum of Ideals

Observe the lattice when we collapse normal subgroups down to 0.

$$\frac{\langle p \rangle}{\langle g(X) \rangle} \subseteq \frac{\mathbb{Z}[X]}{\langle g(X) \rangle} \Leftrightarrow \langle p \rangle \subseteq \mathbb{Z}[X]$$

$$\phi : \mathbb{Z}[X]/\langle g(X) \rangle \to \mathbb{Z}[X]/\langle p, g(X) \rangle$$
$$\phi(r + \langle g(X) \rangle) = r + \langle p, g(X) \rangle$$
$$\ker \phi = \langle p, g(X) \rangle$$

Then observe

$$\phi(r + \langle g(X) \rangle) = 0 \Leftrightarrow r \in \langle p, g(X) \rangle \Leftrightarrow r + \langle g(X) \rangle \in \langle p, g(X) \rangle$$

By first iso theorem with the homomorphism ϕ, we see that

$$(\mathbb{Z}[X]/\langle g(X) \rangle)/\langle p, g(X) \rangle \cong \mathbb{Z}[X]/\langle p, g(X) \rangle$$

Alternatively we can observe that $\langle g(X) \rangle \subseteq \langle p, g(X) \rangle \subseteq \mathbb{Z}[X]$, and then by the third theorem

$$\frac{\mathbb{Z}[X]/\langle g(X) \rangle}{\langle p, g(X) \rangle/\langle g(X) \rangle} \cong \frac{\mathbb{Z}[X]}{\langle p, g(X) \rangle}$$

since $\langle p, g(X) \rangle/\langle g(X) \rangle = \langle p, g(X) \rangle$.

$\mathbb{Z}_K/\mathfrak{p}_1 \cong \mathbb{F}_p[X]/\langle \bar{g}_1(X)\rangle$ and is a Field

$$\mathbb{Z}_K/\mathfrak{p}_1 = \mathbb{Z}[\gamma]/\langle 5, \gamma^2+\gamma+1\rangle$$

$$\phi : \mathbb{Z}[\gamma] \to \mathbb{Z}[X]/\langle g(X)\rangle$$
$$\phi(a_0 + a_1\gamma + a_2\gamma^2 + a_3\gamma^3) = a_0 + a_1X + a_2X^2 + a_3X^3 + \langle g(X)\rangle$$

$$\frac{\mathbb{Z}[\gamma]}{\langle p, g_1(\gamma)\rangle} \cong \frac{\mathbb{Z}[X]/\langle g(X)\rangle}{\langle p, g_1(X), g(X)\rangle/\langle g(X)\rangle} \cong \frac{\mathbb{Z}[X]}{\langle p, g_1(X), g(X)\rangle}$$

But also going in reverse with $\psi : \mathbb{Z}[X]/\langle p\rangle \to \mathbb{F}_p$

$$\frac{\mathbb{Z}[X]}{\langle p, g_1(X), g(X)\rangle} \cong \frac{\mathbb{Z}[X]/\langle p\rangle}{\langle p, g_1(X), g(X)\rangle/\langle p\rangle} \cong \frac{\mathbb{F}_p[X]}{\langle \bar{g}_1(X), \bar{g}(X)\rangle}$$

Note that $\bar{g}_1(X)|\bar{g}(X)$

$$\mathbb{Z}_K/\mathfrak{p}_1 \cong \mathbb{F}_p[X]/\langle \bar{g}_1(X)\rangle$$

$\bar{g}_1(X)$ is irreducible $\Rightarrow \langle \bar{g}_1(X)\rangle$ is a prime ideal $\Rightarrow$ the right hand side is a field, and so $\mathfrak{p}_1$ is a prime ideal.

$\mathbb{Z}_K/\langle p\rangle \cong \mathbb{F}_p[X]/\langle \bar{g}(X)\rangle$

$$\begin{aligned}
\mathbb{Z}_K/\langle p\rangle &= \mathbb{Z}[\gamma]/\langle p\rangle \\
&\cong \frac{\mathbb{Z}[X]/\langle g(X)\rangle}{\langle p, g(X)\rangle/\langle g(X)\rangle} \\
&= \frac{\mathbb{Z}[X]}{\langle p, g(X)\rangle} \\
&\cong \frac{\mathbb{Z}[X]/\langle p\rangle}{\langle p, g(X)\rangle/\langle p\rangle}
\end{aligned}$$

But let $r \in \langle p, g(X)\rangle/\langle p\rangle \subseteq \mathbb{Z}[X]/\langle p\rangle$, then $r = ap + bg(X) \in \langle p, g(X)\rangle + \langle p\rangle = \langle p, g(X)\rangle$

$$\begin{aligned}
\frac{\mathbb{Z}[X]/\langle p\rangle}{\langle p, g(X)\rangle/\langle p\rangle} &= \frac{\mathbb{Z}[X]/\langle p\rangle}{\langle p, g(X)\rangle} \\
&\cong \frac{\mathbb{F}_p[X]}{\langle \bar{g}(X)\rangle} \\
&\cong \mathbb{Z}_K/\langle p\rangle
\end{aligned}$$

Deconstructing $p\mathbb{Z}_K$

There is a map $\mathbb{Z}_K \to \mathbb{Z}_K/\langle p\rangle$ with kernel $\langle p\rangle$.

Then for each component of the decomposed $\mathbb{Z}_K/\langle p\rangle$, there is another map $\mathbb{Z}_K/\langle p\rangle \to \mathbb{Z}_K/\mathfrak{p}_1 \cong \mathbb{F}_p[X]/\langle \bar{g}_1(X)\rangle$ by $\gamma \to X \mod \langle p, g_1(X)\rangle$. So the kernel is $\langle p, g_1(\gamma)\rangle$.

$$p\mathbb{Z}_K = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}$$